



Statistics of Financial Markets

Chapter 4: Probability



Daniel Traian PELE

Bucharest University of Economic Studies

IDA Institute Digital Assets

Blockchain Research Center

AI4EFin Artificial Intelligence for Energy Finance

Romanian Academy, Institute for Economic Forecasting

MSCA Digital Finance

Acronyms used in this material

- AI: Artificial Intelligence
- AR: AutoRegressive (model)
- ARCH: AutoRegressive Conditional Heteroskedasticity
- BET: Bucharest Exchange Trading (index)
- CDF: Cumulative Distribution Function
- CLT: Central Limit Theorem
- CRR: Cox–Ross–Rubinstein (binomial model)
- ES: Expected Shortfall
- GARCH: Generalised AutoRegressive Conditional Heteroskedasticity
- GBM: Gradient Boosting Machine
- IBM: International Business Machines (company)
- LCG: Linear Congruential Generator
- LLN: Law of Large Numbers
- PCG64: Permuted Congruential Generator, 64-bit (NumPy default)
- PMF: Probability Mass Function
- RANDU: RANDU (IBM random number generator of the 1960s)
- SFE: Statistics of Financial Markets, Exercises (Quantlet collection of the textbook)

Today's Question and Route

- ▣ **Question:** which probability tools do we need to describe returns and to simulate prices?
 - ▶ every later chapter (tails, volatility, VaR, efficiency) is built on them
- ▣ **Route** of the chapter
 - ▶ random variables, distributions, expectation, variance, covariance
 - ▶ independence vs uncorrelatedness, conditional expectation and conditional variance
 - ▶ random numbers and Monte Carlo; the binomial model
 - ▶ random walk, martingale, white noise, AR(1); Wiener process and GBM
- ▣ Real data, 2000–2026: S&P 500, DAX, BET and Bitcoin (from 2014)

Learning Outcomes

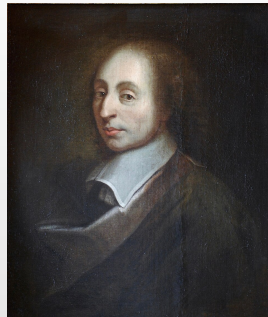
- Compute expectations, variances, covariances and correlations, and the volatility of a portfolio
- Explain why uncorrelated returns can still be dependent, and measure it
- Use conditional expectation and conditional variance, the building blocks of GARCH
- Generate random numbers with the inverse transform and estimate a probability by Monte Carlo, with its standard error
- Define a random walk, a martingale, white noise, the Wiener process and GBM, and simulate them

Reading and Tools

- Textbook: [Franke, Härdle & Hafner \(2019\)](#), *Statistics of Financial Markets*, 5th ed., Ch. 3 (Sec. 3.1–3.5) and Ch. 4 (discrete-time stochastic processes)
 - ▶ exercises with solutions: [Borak, Härdle & López-Cabrera \(2013\)](#), Ch. 3–4
- Monte Carlo methods in finance: [Glasserman \(2003\)](#)
- Python Quantlets of this chapter: [Quantlets/Ch_04](#)
 - ▶ ported from the SFE Quantlets of the textbook (SFErandu, SFErangen1, SFErangen2, SFEBinomp, SFEWienerProcess, SFEsimGBM)
 - ▶ each chart in these slides links to the Quantlet that draws it
- Lecture notebook: [open in Google Colab](#)
- Video course:  [Statistics of Financial Markets](#)

1654: the Problem of Points

- Two players stop a fair game before the end: how should the stake be split?
- Blaise Pascal and Pierre de Fermat solved it in letters (1654): split by the probability of winning, not by the score
- Example: A needs 1 more win, B needs 2; at most two more rounds
 - ▶ B wins only if B wins both rounds: probability $1/4$
 - ▶ A should get 75% of the stake: its **expected** gain
- The same idea prices a derivative today: a fair price is an expected payoff

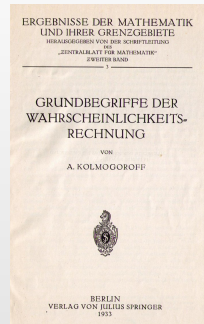


Blaise Pascal (1623–1662)

Portrait: unknown author, after François II Quesnel (c. 1690); public domain; Wikimedia Commons

Kolmogorov's Axioms (1933)

- A **probability space** $(\Omega, \mathcal{F}, P)$ Kolmogoroff (1933)
 - ▶ Ω : all possible outcomes (all possible paths of tomorrow's market)
 - ▶ $\mathcal{F}$: the events we can assign a probability to (a σ -algebra)
 - ▶ P : a function from events to $[0, 1]$
- Axioms
 - ▶ $P(A) \geq 0$; $P(\Omega) = 1$
 - ▶ for disjoint $A_1, A_2, \dots$: $P(\cup_i A_i) = \sum_i P(A_i)$
- Everything else (expectation, independence, processes) is built on these three rules



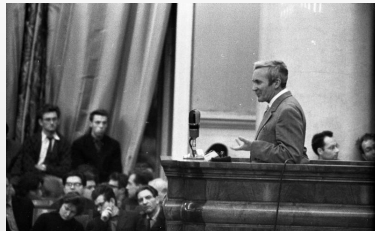
Title page of the *Grundbegriffe*, Springer 1933

Photo: KurtSchwitters (2012); CC BY-SA 3.0;

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Andrey Kolmogorov

- ▣ Andrey Nikolaevich Kolmogorov (1903–1987), Moscow State University
- ▣ Made probability a branch of measure theory: events are sets, probability is a measure
- ▣ Also: conditional expectation in its modern form, Markov processes, turbulence, complexity
- ▣ The Kolmogorov–Smirnov test (Chapter 6) carries his name



Kolmogorov lecturing in Moscow, 1963–1964

Photo: Vsevolod Tarasevich (1963–1964); CC BY 4.0; Wikimedia Commons

Events, Conditional Probability, Independence

- ▣ **Conditional probability:** $P(A | B) = P(A \cap B)/P(B)$, for $P(B) > 0$
- ▣ **Bayes' rule:** $P(B | A) = P(A | B)P(B)/P(A)$
 - ▶ reverses the direction of conditioning: from “signal given crisis” to “crisis given signal”
- ▣ **Independent events:** $P(A \cap B) = P(A)P(B)$, i.e. $P(A | B) = P(A)$
- ▣ S&P 500, 2000–2026: $P(\text{down day}) = 46.3\%$; $P(\text{down} | \text{down yesterday}) = 43.4\%$
 - ▶ two down days in a row: 20.1%, against 21.4% under independence
 - ▶ BET: 50.0% after a down day vs 42.6% after an up day: thin trading makes signs persist

Recap: From Games to Axioms

- ▣ Probability space $(\Omega, \mathcal{F}, P)$ and three axioms
- ▣ Fair value = expected value: the idea of Pascal and Fermat
- ▣ Conditional probability measures how information changes the odds

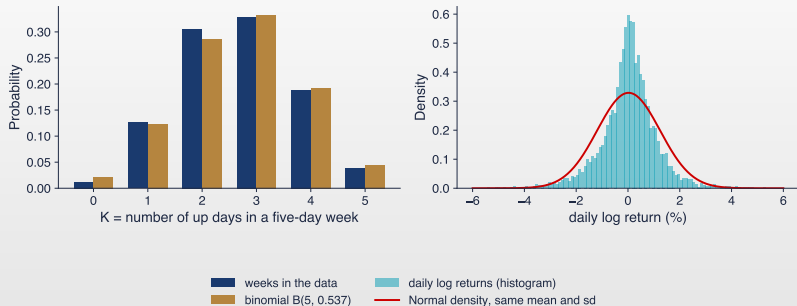
Random Variables

- A **random variable** $X : \Omega \rightarrow \mathbb{R}$ assigns a number to each outcome
 - ▶ tomorrow's log return, the number of up days in a week, the loss of a portfolio
- **Discrete**: finitely or countably many values x_i
 - ▶ **PMF** (probability mass function): $p(x_i) = P(X = x_i)$, with $\sum_i p(x_i) = 1$
- **Continuous**: $P(a < X \leq b) = \int_a^b f(x) dx$
 - ▶ f is the **PDF** (probability density function), $f \geq 0$, $\int f = 1$; $P(X = x) = 0$ for every single x
- Textbook: [Franke, Härdle & Hafner \(2019\)](#), Sec. 3.1

The Distribution Function and the Quantile Function

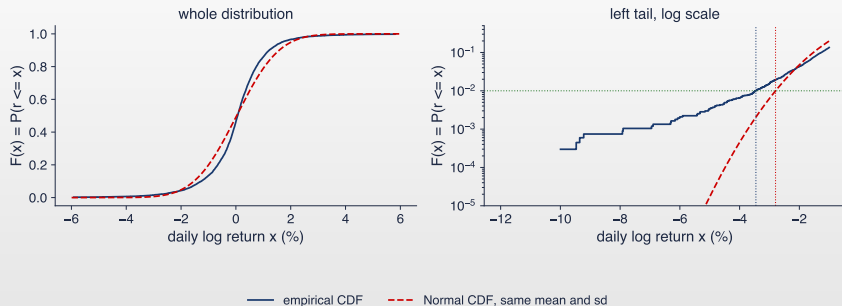
- **CDF** (cumulative distribution function): $F(x) = P(X \leq x)$
 - ▶ non-decreasing, right-continuous, $F(-\infty) = 0$, $F(\infty) = 1$; $f = F'$ for a continuous X
- **Quantile** of order α : $q_\alpha = F^{-1}(\alpha) = \inf\{x : F(x) \geq \alpha\}$
 - ▶ median = $q_{0.5}$; the 1% quantile is the return beaten 99% of the days
- **VaR** (Value at Risk) 1%: $\text{VaR}_{1\%} = -q_{0.01}$, the loss exceeded with probability 1%
 - ▶ a quantile of the return distribution; estimation and backtesting in Chapter 10
- The **empirical CDF** of a sample: $\hat{F}(x) = \frac{1}{n} \#\{t : r_t \leq x\}$

A Discrete and a Continuous Random Variable



- Left: $K = \text{up days in a full week of the S\&P 500 (1 144 weeks)}$ vs $B(5, p)$ with $p = 0.537$: mean 2.67 vs 2.69, variance 1.16 vs 1.24
- Right: daily log returns, a continuous random variable; the histogram estimates the PDF
- The binomial fits K well; the Normal density misses the peak and the tails of the returns ([Chapter 2](#))

Empirical CDF and the 1% Quantile of the S&P 500



- $n = 6714$ days; median 0.063%; $\hat{F}(0) = 46.3\%$ of the days are not up days
- VaR 1%: empirical 3.45%, Normal 2.80%; $P(r < -3\%)$: 1.55% vs 0.63%
- $P(r < -5\%)$: 0.34% in the data vs $1.7 \times 10^{-3}\%$ under the Normal distribution, 199 times less

Recap: Random Variables

- PMF for discrete, PDF for continuous variables; the CDF works for both
- Quantiles invert the CDF; VaR 1% is minus the 1% quantile
- The tails of the empirical CDF are much heavier than the Normal ones

Expectation

- **Expectation** (mean): $E[X] = \sum_i x_i p(x_i)$ or $E[X] = \int x f(x) dx$
 - ▶ the centre of mass of the distribution; it exists only if $E|X| < \infty$ (not for Cauchy, [Chapter 3](#))
- Of a function: $E[g(X)] = \int g(x) f(x) dx$, without finding the distribution of $g(X)$
- **Linearity**: $E[aX + bY + c] = aE[X] + bE[Y] + c$, always (no independence needed)
 - ▶ the expected return of a portfolio is the weighted mean of the expected returns
- In general $E[g(X)] \neq g(E[X])$: $E[e^X] > e^{E[X]}$ (Jensen), the volatility drag of [Chapter 1](#)

Variance and Moments

- **Variance:** $\text{Var}(X) = E[(X - \mu)^2] = E[X^2] - \mu^2$; standard deviation $\sigma = \sqrt{\text{Var}(X)}$ (volatility)
- $\text{Var}(aX + b) = a^2 \text{Var}(X)$: a shift does not change risk, a scale does
- Moments $E[X^k]$; skewness and kurtosis use $k = 3, 4$ ([Chapter 2](#))
- Worked example: S&P 500 daily mean 0.0247%, s.d. 1.213%, 251 days a year
 - ▶ for i.i.d. daily returns (independent and identically distributed): $E[\sum r_t] = 251 \times 0.0247 = 6.2\%$ a year
 - ▶ $\text{Var}(\sum r_t) = 251 \sigma^2$, so annual volatility $1.213\sqrt{251} = 19.2\%$: the square-root-of-time rule

How Rare Is a 4-Sigma Day? Chebyshev's Bound

- ▣ **Chebyshev's inequality:** for any X with finite variance, $P(|X - \mu| \geq k\sigma) \leq 1/k^2$
 - ▶ it uses only the mean and the variance, so it holds for every distribution
- ▣ $k = 4$, S&P 500, 2000–2026
 - ▶ Chebyshev bound: at most 6.25% of the days
 - ▶ Normal distribution: 0.0063%
 - ▶ data: 46 days, 0.69%
- ▣ Reality is between the two: the Normal distribution is far too optimistic, the worst case of Chebyshev too pessimistic

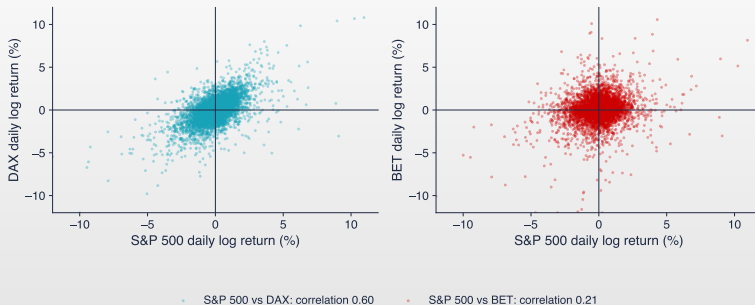
Recap: Expectation and Variance

- ▣ Expectation is linear; variance scales with the square
- ▣ For i.i.d. returns the mean grows with h and the volatility with $\sqrt{h}$
- ▣ Chebyshev bounds tail probabilities for any distribution

Joint Distribution, Covariance, Correlation

- **Joint CDF:** $F(x, y) = P(X \leq x, Y \leq y)$; marginals $F_X(x) = F(x, \infty)$
 - ▶ joint PDF $f(x, y)$; marginal $f_X(x) = \int f(x, y) dy$
- **Covariance:** $\text{Cov}(X, Y) = E[(X - \mu_X)(Y - \mu_Y)] = E[XY] - \mu_X \mu_Y$
- **Correlation:** $\rho = \text{Cov}(X, Y) / (\sigma_X \sigma_Y) \in [-1, 1]$
 - ▶ measures **linear** dependence only; $|\rho| = 1$ iff $Y = a + bX$
- **Variance of a sum:** $\text{Var}(aX + bY) = a^2 \sigma_X^2 + b^2 \sigma_Y^2 + 2ab \text{Cov}(X, Y)$
 - ▶ with weights w and covariance matrix Σ : $\sigma_p^2 = w^\top \Sigma w$

Joint Distributions of Daily Returns



- Common trading days only: prices are joined first, then differenced ($n = 6\,605$ days for S&P 500 and DAX)
- S&P 500 and DAX: $\rho = 0.60$; S&P 500 and BET: $\rho = 0.21$
- The cloud is elliptic in the centre, but extreme days are joint: crises hit all markets together

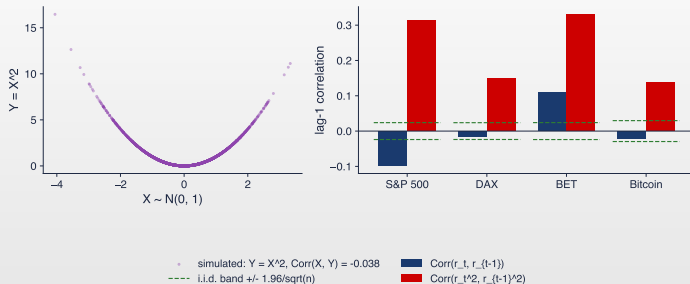
Worked Example: a 50/50 Portfolio of S&P 500 and DAX

- Daily s.d.: $\sigma_1 = 1.22\%$, $\sigma_2 = 1.43\%$; covariance 1.045; correlation 0.60
- $\sigma_p^2 = 0.25\sigma_1^2 + 0.25\sigma_2^2 + 2 \times 0.25 \text{Cov} = 0.883 + 0.522 = 1.405$
 - ▶ $\sigma_p = 1.19\%$, below the average volatility 1.32%: diversification
- S&P 500 and BET, lower correlation: $\sigma_p = 1.04\%$ vs average 1.33%, a larger gain
- Diversification works only through $\rho < 1$; in a crash correlations rise, exactly when it is needed

Independence and Uncorrelatedness

- X and Y are **independent** if $F(x, y) = F_X(x)F_Y(y)$ for all x, y
 - ▶ then $E[g(X)h(Y)] = E[g(X)]E[h(Y)]$ for **all** functions g, h
- Independent $\Rightarrow$ uncorrelated ($g, h = \text{identity}$); the converse is false
- Example: $X \sim N(0, 1)$, $Y = X^2$
 - ▶ $\text{Cov}(X, X^2) = E[X^3] = 0$: uncorrelated
 - ▶ but Y is a function of X : completely dependent
- Exception: for jointly Normal variables, uncorrelated $\Leftrightarrow$ independent

Returns Are Uncorrelated, Squared Returns Are Not



- Left: simulated $Y = X^2$, sample correlation -0.038 , yet Y is determined by X
- Right: $\text{Corr}(r_t, r_{t-1})$ is small (S&P 500 -0.099 , DAX -0.017 , Bitcoin -0.022); $\text{Corr}(r_t^2, r_{t-1}^2)$ is far outside the band: 0.31, 0.15, 0.14
- BET: $\text{Corr}(r_t, r_{t-1}) = 0.109$, a trace of thin trading; the S&P 500 value comes mostly from crisis days

Recap: Joint Distributions

- ▣ Covariance enters the variance of every portfolio
- ▣ Independent implies uncorrelated, not the other way round
- ▣ Returns: nearly uncorrelated, but their squares are correlated, so they are not independent

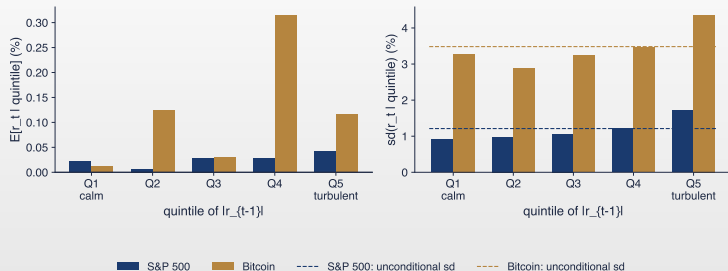
Conditional Expectation

- Conditional density: $f(y | x) = f(x, y)/f_X(x)$
 - ▶ the distribution of Y once we know that $X = x$
- **Conditional expectation:** $E[Y | X = x] = \int yf(y | x) dy$; $E[Y | X]$ is a random variable, a function of X
- It is the best forecast of Y from X : it minimises $E[(Y - g(X))^2]$ over all functions g
 - ▶ for returns: $E[r_t | \mathcal{F}_{t-1}]$, the forecast given all information $\mathcal{F}_{t-1}$ up to yesterday
- **Tower property** (law of iterated expectations): $E[E[Y | X]] = E[Y]$

Conditional Variance and the Law of Total Variance

- **Conditional variance:** $\text{Var}(Y | X) = E[(Y - E[Y | X])^2 | X]$
- **Law of total variance:** $\text{Var}(Y) = E[\text{Var}(Y | X)] + \text{Var}(E[Y | X])$
 - ▶ within-group variance plus between-group variance
- Worked example, two regimes: calm (probability 0.7, mean 0.05%, s.d. 0.8%), turbulent (0.3, -0.10%, 2%)
 - ▶ $E[r] = 0.7 \times 0.05 + 0.3 \times (-0.10) = 0.005\%$
 - ▶ $E[\text{Var}] = 0.7 \times 0.64 + 0.3 \times 4 = 1.648$; $\text{Var}(E) = 0.0047$
 - ▶ $\text{Var}(r) = 1.653$, s.d. 1.286%: almost all risk comes from the changing variance, not from the changing mean

Today's Return Given Yesterday's Move



- ▣ Quintiles of $|r_{t-1}|$: Q1 = the calmest 20% of yesterdays, Q5 = the most turbulent 20%
- ▣ Conditional mean: between 0.006% and 0.042% for the S&P 500, no usable pattern
- ▣ Conditional s.d.: S&P 500 from 0.91% (Q1) to 1.71% (Q5), ratio 1.88; Bitcoin ratio 1.33
- ▣ Law of total variance, S&P 500: $1.469 = 1.469 + 0.00013$; the between part is 0.01%

From Conditional Variance to GARCH

- Write the return as $r_t = \mu_t + \sigma_t z_t$, with z_t i.i.d., mean 0, variance 1
 - ▶ $\mu_t = E[r_t | \mathcal{F}_{t-1}]$: the conditional mean, close to constant
 - ▶ $\sigma_t^2 = \text{Var}(r_t | \mathcal{F}_{t-1})$: the conditional variance, which changes with the news
- ARCH (autoregressive conditional heteroskedasticity) Engle (1982): $\sigma_t^2 = \omega + \alpha r_{t-1}^2$
 - ▶ GARCH Bollerslev (1986): $\sigma_t^2 = \omega + \alpha r_{t-1}^2 + \beta \sigma_{t-1}^2$ (Chapter 9)
- Exactly the pattern in the chart: a large $|r_{t-1}|$ raises today's conditional s.d.
- The unconditional distribution is a mixture over σ_t : heavier tails than the Normal distribution, even if z_t is Normal

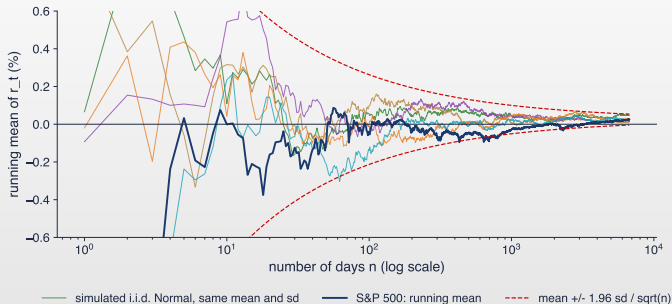
Recap: Conditioning

- ▣ $E[Y | X]$ is the best forecast; the tower property averages it back to $E[Y]$
- ▣ Returns: conditional mean almost flat, conditional variance strongly time-varying
- ▣ Modelling σ_t^2 is the job of ARCH and GARCH

Two Limit Theorems

- ▣ **LLN** (law of large numbers): for i.i.d. X_i with $E|X| < \infty$, $\bar{X}_n \rightarrow \mu$ as $n \rightarrow \infty$
 - ▶ the reason why averages, frequencies and Monte Carlo estimates work
- ▣ **CLT**: with finite variance, $\sqrt{n}(\bar{X}_n - \mu)/\sigma \rightarrow N(0, 1)$ ([Chapter 2](#))
 - ▶ gives the speed: the error of $\bar{X}_n$ is about $\sigma/\sqrt{n}$
 - ▶ infinite variance: sums converge to α -stable laws instead ([Chapter 3](#))
- ▣ Both need some form of independence; dependent returns converge more slowly

The LLN for the Mean Return: Slow



- ▣ S&P 500, 26.7 years: mean 0.0247% a day, standard error $\sigma / \sqrt{n} = 0.0148\%$, $t = 1.67$
- ▣ Annual mean 6.2%, 95% interval $[-1.1\%, 13.5\%]$: even 26 years do not pin down the expected return
- ▣ For $t = 3$ at this mean and volatility we would need about 86 years of data

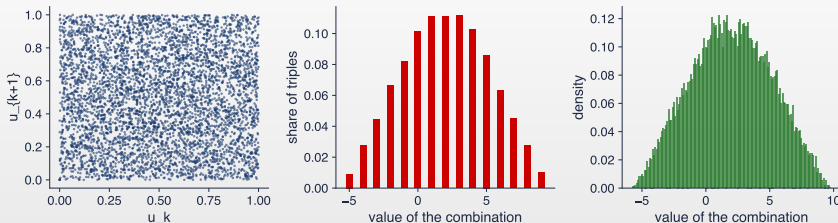
The Monte Carlo Idea

- Estimate $\theta = E[g(X)]$ by $\hat{\theta}_N = \frac{1}{N} \sum_{i=1}^N g(X_i)$ with simulated X_i
 - ▶ a probability is an expectation: $P(A) = E[1_A]$
 - ▶ the LLN makes $\hat{\theta}_N \rightarrow \theta$; the CLT gives the error $\text{sd}(g(X))/\sqrt{N}$
- Named after the Monaco casino by Ulam and von Neumann, Los Alamos, 1940s [Metropolis & Ulam \(1949\)](#)
- In finance [Glasserman \(2003\)](#)
 - ▶ prices of path-dependent options, VaR and ES of portfolios, stress scenarios
 - ▶ any expectation without a closed form
- Two ingredients: uniform random numbers and a way to turn them into any distribution

Pseudo-Random Numbers: Linear Congruential Generators

- **LCG** (linear congruential generator): $x_{k+1} = (ax_k + c) \bmod M$, $u_k = x_k/M \in [0, 1)$
 - ▶ deterministic: the same seed x_0 gives the same sequence, which makes results reproducible
- Worked example (SFERangen1): $a = 5$, $c = 1$, $M = 16$, $x_0 = 7$
 - ▶ x : 7, 4, 5, 10, 3, 0, 1, 6, 15, 12, 13, 2, 11, 8, 9, 14, 7
 - ▶ period 16, the maximum M ; with $a = 4$ the sequence sticks after one step (period 1)
- Real generators use huge M ; the period must be far longer than any simulation
- Good uniform marginals are not enough: consecutive numbers must also look independent

RANDU: Random Numbers Fall Mainly in the Planes (SFERandu)



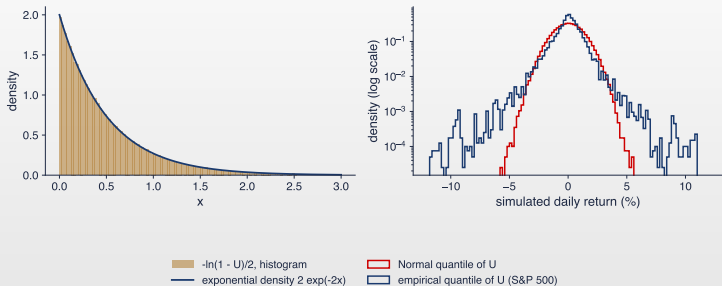
• RANDU pairs (u_k, u_{k+1}) ■ RANDU: $9u_k - 6u_{k+1} + u_{k+2}$ ■ PCG64: the same combination

- RANDU, IBM, 1960s: $x_{k+1} = 65539 x_k \bmod 2^{31}$; pairs look uniform (left), correlation 0.003
- But $9u_k - 6u_{k+1} + u_{k+2}$ takes only 15 integer values, from -5 to 9: all triples lie on 15 planes [Marsaglia \(1968\)](#)
- Modern generators: Mersenne Twister [Matsumoto & Nishimura \(1998\)](#), PCG64 (the NumPy default, right panel)

The Inverse Transform Method

- **Theorem:** if $U \sim U(0, 1)$ and F is a CDF, then $X = F^{-1}(U)$ has CDF F
 - ▶ proof: $P(X \leq x) = P(F^{-1}(U) \leq x) = P(U \leq F(x)) = F(x)$
- Exponential with rate λ : $F(x) = 1 - e^{-\lambda x}$, so $X = -\ln(1 - U)/\lambda$
 - ▶ $\lambda = 2$, $u = 0.9$: $x = -\ln(0.1)/2 = 1.151$
- Normal: $X = \mu + \sigma \Phi^{-1}(U)$
- Data: use the empirical quantile function $\hat{F}^{-1}$
 - ▶ this is resampling past days: historical simulation and the bootstrap

Inverse Transform: Exponential, Normal and Historical

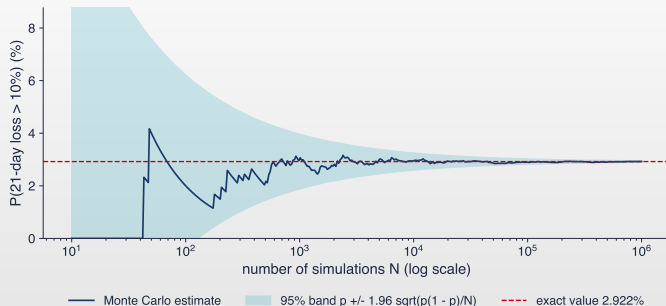


- Left: 200 000 draws — $-\ln(1 - U)/2$ match the exponential density; sample mean 0.500 (theory $1/\lambda = 0.5$)
- Right: the same uniforms through the Normal and the empirical S&P 500 quantile functions
- Draws below -5% : 0.0035% Normal vs 0.34% historical (data: 0.34%); worst draw -5.7% vs -12.7%

Worked Example: a Monthly Loss Probability by Monte Carlo

- Model: S&P 500 daily log returns i.i.d. $N(0.0247, 1.213^2)$ (in %); event: 21-day log return below -10%
- Exact: the sum is $N(0.519, 5.56^2)$, so $p = \Phi(-1.892) = 2.922\%$
- Monte Carlo: $\hat{p}_N$ = share of simulated months below -10% ; standard error $\sqrt{p(1-p)/N}$
 - ▶ $N = 1\,000$: 3.000% (s.e. 0.533%); $N = 100\,000$: 2.901% (s.e. 0.053%)
 - ▶ 10 times more precision needs 100 times more simulations
- For a relative error of 10% (95% level) we need $N \approx 1.96^2(1-p)/(0.1^2p) = 12\,762$

Monte Carlo Converges at Rate $1/\sqrt{N}$



- ▣ The estimate wanders for small N and settles inside the 95% band, which shrinks like $1/\sqrt{N}$
- ▣ $N = 10^6$: 2.926% vs exact 2.922%, s.e. 0.017%
- ▣ Monte Carlo removes the simulation error, not the model error: Seminar 4 (B3) repeats it with resampled days

Recap: Random Numbers and Monte Carlo

- ▣ Pseudo-random generators are deterministic; check them in several dimensions
- ▣ $F^{-1}(U)$ turns uniforms into any distribution, including the empirical one
- ▣ Monte Carlo error $\propto 1/\sqrt{N}$; always report it

The Binomial Process

- Each period the price moves up by a factor u with probability p or down by $d < u$ with probability $1 - p$, independently
 - ▶ after n steps with K up moves: $S_n = S_0 u^K d^{n-K}$, $K \sim B(n, p)$
- $\ln S_n = \ln S_0 + K \ln u + (n - K) \ln d$: a random walk in log prices (Section 9)
- $E[S_n] = S_0 (pu + (1 - p)d)^n$, because the steps are independent
 - ▶ the basis of the binomial option pricing model [Cox, Ross & Rubinstein \(1979\)](#); textbook: [Franke, Härdle & Hafner \(2019\)](#), Ch. 7

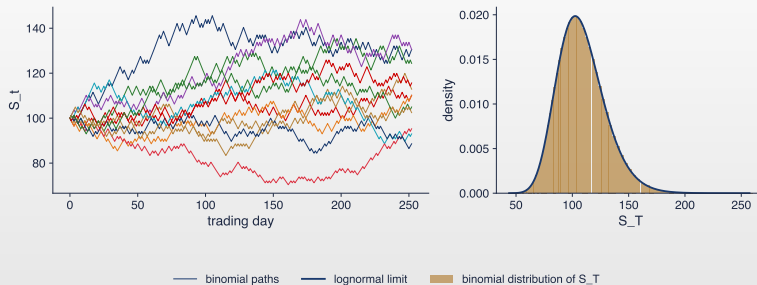
Worked Example: a Two-Step Tree

- $S_0 = 100$, $u = 1.2$, $d = 0.8$, $p = 0.6$
- Terminal values and probabilities
 - ▶ uu : 144 with $p^2 = 0.36$; ud or du : 96 with $2p(1-p) = 0.48$; dd : 64 with $(1-p)^2 = 0.16$
- $E[S_2] = 100 \times (0.6 \times 1.2 + 0.4 \times 0.8)^2 = 100 \times 1.04^2 = 108.16$
- $P(S_2 < 100) = 0.48 + 0.16$: an up and a down move lose money, because $ud = 0.96 < 1$
- With $p^* = (1-d)/(u-d) = 0.50$ the price is a fair game, $E[S_{t+1} | S_t] = S_t$ (Section 9)

Calibrating the Tree to a Market (Cox–Ross–Rubinstein)

- **CRR** (Cox–Ross–Rubinstein) with step Δt : $u = e^{\sigma\sqrt{\Delta t}}$, $d = 1/u$, $p = (e^{\mu\Delta t} - d)/(u - d)$
 - ▶ so that the mean and the variance of each step match a market with drift μ and volatility σ
- S&P 500, 2000–2026: $\sigma = 19.23\%$, $\mu = 8.06\%$ a year; daily steps, $\Delta t = 1/252$
 - ▶ $u = 1.0122$, $d = 0.9880$, $p = 0.5102$
- As $\Delta t \rightarrow 0$, $\ln S_T$ becomes Normal (CLT for K): the tree converges to GBM (Section 10)

Binomial Paths and the Lognormal Limit (SFEBinomp)



- Left: 12 paths over one year of daily steps, $S_0 = 100$
- Right: distribution of S_T with 252 steps; $E[S_T] = 108.40 = 100e^\mu$; 5% quantile 78.5 vs lognormal 77.6
- $P(S_T < 100)$: 39.7% binomial vs 37.3% lognormal ($S_T = 100$ exactly has a probability mass, which the binomial counts on one side)

Recap: The Binomial Model

- Up or down each period: $S_n = S_0 u^K d^{n-K}$ with binomial K
- CRR matches μ and σ ; the limit is lognormal
- A special p^* turns the price into a fair game: the key to option pricing

Stochastic Processes and Information

- A **stochastic process** $\{X_t\}$: a family of random variables indexed by time
 - ▶ one realisation = one path; a price series is a single path of the process
- $\mathcal{F}_t$: the information available at time t (past prices, news); it grows with t (a **filtration**)
- **(Weakly) stationary**: $E[X_t]$ and $\text{Cov}(X_t, X_{t+h})$ do not depend on t
 - ▶ returns: roughly yes; prices: no
- Textbook: [Franke, Härdle & Hafner \(2019\)](#), Ch. 4 and Sec. 11.3

White Noise and the Random Walk

- **White noise:** $E[\varepsilon_t] = 0$, $\text{Var}(\varepsilon_t) = \sigma^2$, $\text{Cov}(\varepsilon_t, \varepsilon_s) = 0$ for $t \neq s$
 - ▶ i.i.d. noise is stronger: the ε_t are independent, not only uncorrelated
- **Random walk:** $S_t = S_{t-1} + \mu + \varepsilon_t = S_0 + \mu t + \sum_{s=1}^t \varepsilon_s$
 - ▶ $E[S_t] = S_0 + \mu t$; $\text{Var}(S_t) = t\sigma^2$: not stationary
 - ▶ simulated with $\sigma = 1$: variance 10.1, 49.8, 100.9 at $t = 10, 50, 100$
- Log prices as a random walk = returns as white noise: the benchmark of Chapter 7 (efficient markets)

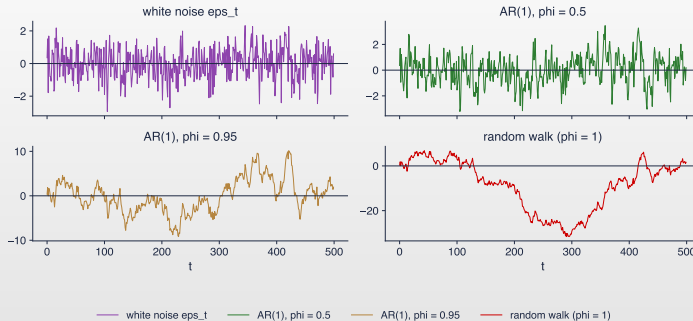
Martingales

- $\{X_t\}$ is a **martingale** if $E|X_t| < \infty$ and $E[X_{t+1} | \mathcal{F}_t] = X_t$
 - ▶ a fair game: the best forecast of tomorrow is today
- A **martingale difference**: $\varepsilon_t = X_t - X_{t-1}$ with $E[\varepsilon_t | \mathcal{F}_{t-1}] = 0$
- Examples
 - ▶ a random walk without drift; $S_t^2 - t\sigma^2$ for the same walk; the binomial price under p^*
- Martingale $\neq$ i.i.d. increments: the conditional **variance** may change, as in GARCH
 - ▶ the efficient-market idea in its weak form [Fama \(1970\)](#): excess returns are a martingale difference

The AR(1) Process

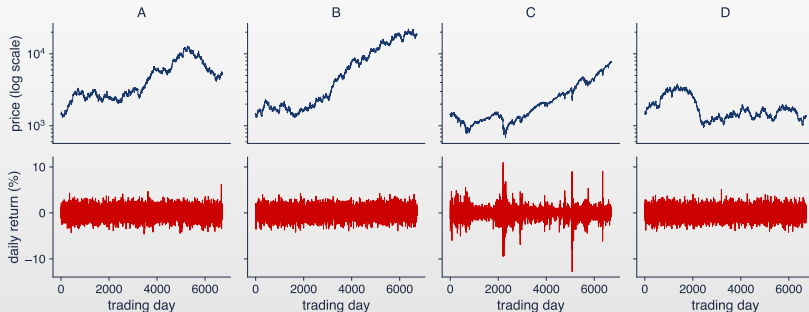
- **AR(1)** (autoregressive of order 1): $X_t = c + \phi X_{t-1} + \varepsilon_t$, ε_t white noise
 - ▶ $\phi = 0$: white noise; $\phi = 1$, $c = \mu$: random walk
- Stationary iff $|\phi| < 1$; then
 - ▶ mean $c/(1 - \phi)$, variance $\sigma^2/(1 - \phi^2)$, autocorrelation $\rho(h) = \phi^h$
 - ▶ half-life of a shock: $\ln 0.5 / \ln \phi$ periods
- Simulated, $\sigma = 1$
 - ▶ $\phi = 0.5$: variance 1.34 (theory 1.33); $\rho(5) = 0.034$ (theory 0.031); half-life 1.0
 - ▶ $\phi = 0.95$: variance 10.20 (theory 10.26); $\rho(5) = 0.772$; half-life 13.5
- Uses: interest rates and spreads (mean reversion); the squared returns behave like a persistent AR, which leads to GARCH

White Noise, AR(1) and Random Walk, Same Shocks



- The larger ϕ , the longer a shock lives: the AR(1) with $\phi = 0.95$ wanders, but returns to 0
- The random walk never returns on purpose: shocks are permanent and its variance grows with t

Question for the Room: Which One Is Real?



- One column is the S&P 500, 2000–2026; three are GBM paths (i.i.d. Normal returns) with the same mean and volatility
- **What do you think?** Which column is real, from the prices alone?
- **What do you think?** Which column is real, from the returns?

Answer: Column C

- **Answer:** column C is the S&P 500
- Prices: hard to tell; random walks with drift produce booms and long falls too
 - ▶ this is why [Bachelier \(1900\)](#) and [Osborne \(1959\)](#) modelled prices as random walks
- Returns: obvious; the real series has quiet years and violent clusters (2008, 2020)
 - ▶ excess kurtosis 10.7 vs at most 0.02 in the simulations; $\text{Corr}(r_t^2, r_{t-1}^2) = 0.31$ vs at most 0.006
 - ▶ largest absolute daily move 12.8% vs 6.2%
- Lesson: a model can fit the price path and still be wrong about risk

Recap: Discrete-Time Processes

- White noise: uncorrelated; i.i.d. noise: independent; a martingale difference: unpredictable mean
- Random walk: permanent shocks, variance $t\sigma^2$; AR(1) with $|\phi| < 1$: shocks fade at rate ϕ^h
- Prices look like random walks; returns are not i.i.d.

From Pollen to Prices: Brownian Motion

- 1827: the botanist Robert Brown sees pollen particles jiggling in water
- 1900: Louis Bachelier models the Paris bond market with the same motion, five years before physics [Bachelier \(1900\)](#)
- 1905: Albert Einstein explains it by molecular collisions [Einstein \(1905\)](#)
- 1923: Norbert Wiener builds the rigorous mathematical object [Wiener \(1923\)](#)
- 1959: [Osborne \(1959\)](#) applies it to log prices: geometric Brownian motion



Robert Brown (1773–1858)

Photo: [Mauil & Polyblank \(1855\)](#); public domain;
[Wikimedia Commons](#)

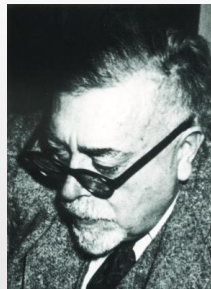


Louis Bachelier (1870–1946)

Photo: [unknown author](#); public domain; [Wikimedia Commons](#)

The Wiener Process

- $\{W(t), t \geq 0\}$ is a **Wiener process** (standard Brownian motion) if
 - ▶ $W(0) = 0$
 - ▶ increments over disjoint intervals are independent
 - ▶ $W(t) - W(s) \sim N(0, t - s)$ for $s < t$
 - ▶ the paths are continuous
- It is the limit of the scaled random walk $W_n(t) = (\varepsilon_1 + \dots + \varepsilon_{\lfloor nt \rfloor}) / \sqrt{n}$, $\varepsilon = \pm 1$ (Donsker)
- Textbook: [Franke, Härdle & Hafner \(2019\)](#), Ch. 5

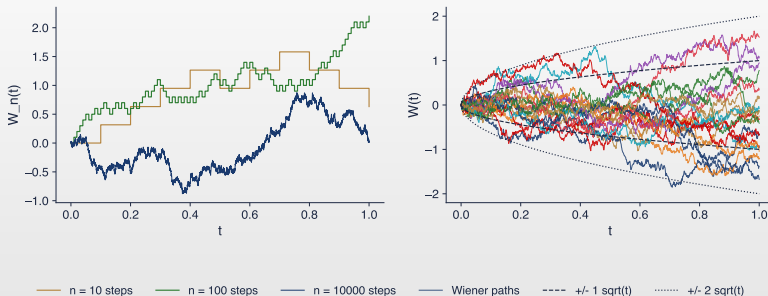


Norbert Wiener (1894–1964)

Photo: [Konrad Jacobs](#); CC BY-SA 2.0 DE;

[Wikimedia Commons](#)

From Random Walks to the Wiener Process (SFEWienerProcess)



- Left: scaled random walks with 10, 100 and 10 000 steps; the steps vanish, the randomness stays
- Right: 20 Wiener paths; $\text{Var}(W(t)) = t$, so they spread like $\pm \sqrt{t}$; simulated $\text{Var}(W(1)) = 1.006$; 68.2% of $W(1)$ inside ± 1 (theory 68.3%)

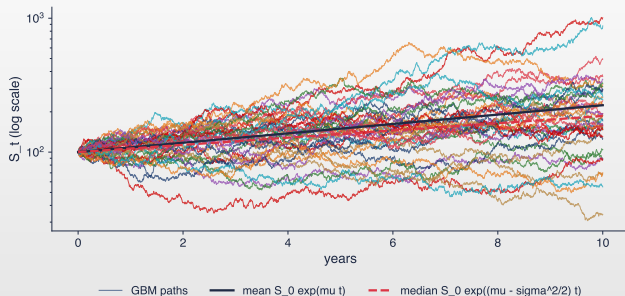
Strange Properties of Wiener Paths

- Continuous everywhere, differentiable nowhere
 - ▶ $(W(t+h) - W(t))/h \sim N(0, 1/h)$ explodes as $h \rightarrow 0$
- **Quadratic variation:** $\sum_k (W(t_{k+1}) - W(t_k))^2 \rightarrow t$ on a finer and finer grid
 - ▶ 1000 steps on $[0, 1]$: mean 0.999, s.d. 0.045 across 10 000 paths
 - ▶ the reason why $(dW)^2 = dt$ in Itô calculus, and why realised variance measures volatility (Chapter 8)
- $\text{Cov}(W(s), W(t)) = \min(s, t)$: simulated $\text{Cov}(W(0.5), W(1)) = 0.496$
- W is a martingale; so is $W(t)^2 - t$

Geometric Brownian Motion

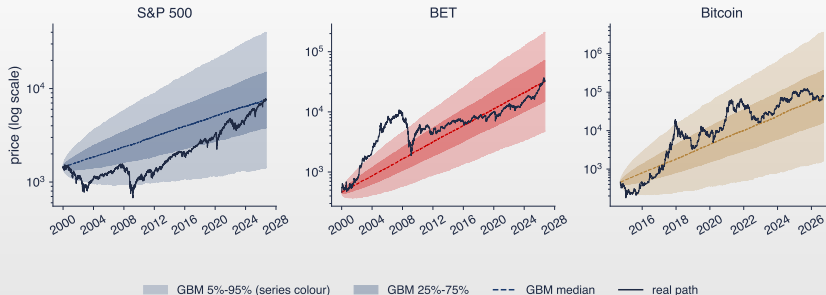
- **GBM** (geometric Brownian motion): $dS_t = \mu S_t dt + \sigma S_t dW_t$
 - ▶ the relative change dS/S has drift μ and volatility σ
- Solution: $S_t = S_0 \exp((\mu - \sigma^2/2)t + \sigma W_t)$
 - ▶ $\ln(S_t/S_0) \sim N((\mu - \sigma^2/2)t, \sigma^2 t)$: lognormal prices, i.i.d. Normal log returns
- Mean $E[S_t] = S_0 e^{\mu t}$; median $S_0 e^{(\mu - \sigma^2/2)t}$: the gap is the volatility drag
- Exact simulation on a grid: $S_{t+\Delta t} = S_t \exp((\mu - \sigma^2/2)\Delta t + \sigma\sqrt{\Delta t} Z)$, $Z \sim N(0, 1)$
- The model behind [Black & Scholes \(1973\)](#) (Chapter 6 of the textbook)

GBM Calibrated to the S&P 500 (SFEsimGBM)



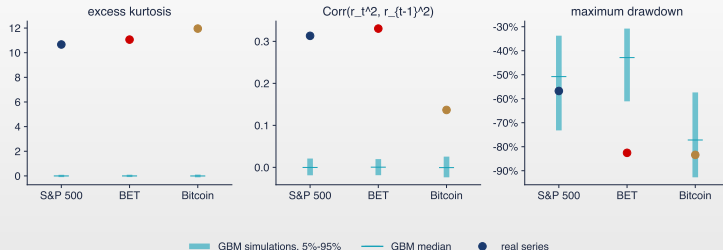
- Calibration, 2000–2026: $\sigma = 19.23\%$, log drift $\mu - \sigma^2/2 = 6.21\%$, so $\mu = 8.06\%$ a year
- After 10 years: mean 224.0, median 186.2; 62.0% of the paths end below the mean (theory $\Phi(\sigma\sqrt{T}/2) = 61.9\%$)
- Probability of a loss after 10 years: 15.3%

Real Paths in the GBM Fan



- Each fan: 4000 GBM paths with the drift and volatility of the series itself, started on its first day
- S&P 500: the path reached the 0.5% quantile on 23 July 2002, then the median at the end; outside the 90% band 7% of the time
- BET: outside the band 24% of the time; the 2000–2007 boom and the 2008 crash are not GBM-like; Bitcoin: $\sigma = 66.6\%$ a year makes the fan enormous

What GBM Misses

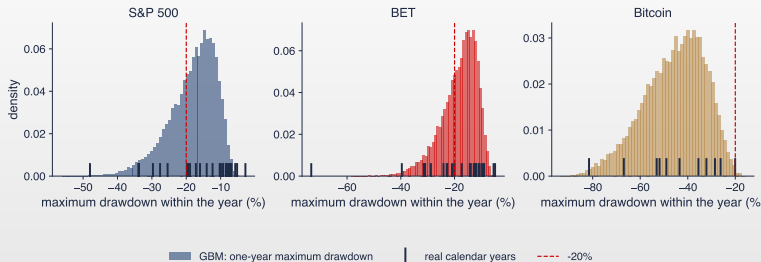


- Excess kurtosis: S&P 500 10.7, BET 11.1, Bitcoin 12.0; 90% of 300 GBM simulations lie within $[-0.10, 0.09]$
- Volatility clustering: $\text{Corr}(r_t^2, r_{t-1}^2) = 0.31, 0.33, 0.14$ vs at most 0.03 under GBM
- Maximum drawdown: S&P 500 -57% is ordinary for GBM; BET -83% is beyond all but 5% of the paths (5% quantile -61%)
- 4-sigma days: 46 (S&P 500) vs at most 1 in 95% of the GBM paths; crypto has the same problem [Pele et al. \(2023\)](#)

Recap: Wiener Process and GBM

- Wiener process: independent Normal increments with variance equal to the elapsed time
- GBM: lognormal prices, median below mean, the benchmark of option pricing
- Real returns have heavy tails and volatility clusters that GBM cannot produce

An Open Question: Does GBM Get Drawdown Risk Right?



- Drawdown within a calendar year beyond 20%: S&P 500 in 6 of 26 years (23%) vs 34% under GBM; BET 46% vs 34%
- Beyond 40%: GBM expects 0.14 such years for the S&P 500 and 0.17 for the BET; both had 1 (2008: -48% and -73%)
- Open: does GBM overstate moderate drawdowns and understate extreme ones because calm and crisis years cluster?

How AI Could Help

- **Literature:** find studies of drawdown distributions and of their link to volatility clustering, and summarise them
- **Code:** draft simulations of yearly drawdowns under GBM, the i.i.d. bootstrap and a GARCH model
- **Design:** propose a test that compares the observed number of bad years with the model probability
- **Example prompt**
 - ▶ Write Python code that simulates 10,000 one-year paths of daily S&P 500 returns under GBM and under an i.i.d. bootstrap of 2000-2026 returns, and returns the probability of a drawdown beyond 20% and 40% within the year.

What to Check

- ▣ Definitions: drawdown from the running peak within the year, not from the first day of the year
- ▣ Calibration: the same μ and σ , the same number of trading days a year as the data
- ▣ Inference: 26 years is a small sample; compute a binomial p -value, not only shares
- ▣ Mechanism: the i.i.d. bootstrap keeps heavy tails but removes clustering; the difference isolates the effect of clustering
- ▣ References: every cited paper must exist; check the DOI

Project Seed

- **Question:** how wrong is GBM about the probability of a bad year, and why?
 - ▶ data: S&P 500, DAX, BET and Bitcoin from the course data
- Steps
 - ▶ compute within-year maximum drawdowns for every calendar year
 - ▶ simulate the same statistic under GBM, the i.i.d. bootstrap and a GARCH(1,1) model (Chapter 9)
 - ▶ compare the shares beyond 20% and 40% with binomial tests
- Deliverable: one table, one chart, and a paragraph on what the data can and cannot show
- Declare any AI use, and list the errors of the AI that you corrected [Wang et al. \(2023\)](#)

Key Takeaways

- ▣ Distributions: PMF, PDF, CDF and quantiles; VaR 1% is minus a quantile
- ▣ Covariance drives portfolio risk; uncorrelated returns are not independent
- ▣ Conditional variance changes with yesterday's news: the starting point of GARCH
- ▣ Monte Carlo: inverse transform plus the LLN; error $\propto 1/\sqrt{N}$
- ▣ Binomial tree $\rightarrow$ GBM; random walk, martingale, white noise, AR(1)
- ▣ GBM fits price paths roughly, but misses heavy tails, clustering and extreme drawdowns

Key Formulas

Quantity	Formula
Variance of a sum	$\text{Var}(aX + bY) = a^2\sigma_X^2 + b^2\sigma_Y^2 + 2ab\text{Cov}(X, Y)$
Correlation	$\rho = \text{Cov}(X, Y)/(\sigma_X\sigma_Y)$
Tower property	$E[E[Y X]] = E[Y]$
Total variance	$\text{Var}(Y) = E[\text{Var}(Y X)] + \text{Var}(E[Y X])$
Inverse transform	$X = F^{-1}(U), U \sim U(0, 1)$
Monte Carlo error	$\sqrt{p(1-p)/N}$
Random walk	$S_t = S_{t-1} + \mu + \varepsilon_t, \text{Var}(S_t) = t\sigma^2$
AR(1)	$\text{Var}(X_t) = \sigma^2/(1 - \phi^2), \rho(h) = \phi^h$
GBM	$S_t = S_0 \exp((\mu - \sigma^2/2)t + \sigma W_t), E[S_t] = S_0 e^{\mu t}$

Check Yourself

- **Question:** $\sigma_1 = 2\%$, $\sigma_2 = 3\%$, $\rho = 0.5$; what is the volatility of the 50/50 portfolio?
 - ▶ **Answer:** $\sqrt{0.25 \times 4 + 0.25 \times 9 + 2 \times 0.25 \times 0.5 \times 2 \times 3} = \sqrt{4.75} = 2.18\%$
- **Question:** GBM with $\mu = 8\%$, $\sigma = 20\%$, $S_0 = 100$; what are the mean and the median of S_1 ?
 - ▶ **Answer:** mean $100e^{0.08} = 108.33$, median $100e^{0.06} = 106.18$
- **Question:** AR(1) with $\phi = 0.8$ and $\sigma = 1$; what is its variance?
 - ▶ **Answer:** $1/(1 - 0.64) = 2.78$
- Next: Chapter 5, heavy tails and extreme value theory

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